Marcelo De La Torre Esquinca

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PROFESSIONAL SUMMARY

Robotics and Digital Systems engineer with international research experience at AIST (Japan) in humanoid robot teleoperation. Skilled in ROS2, embedded systems (STM32, ESP32, Raspberry Pi), real-time control, and probabilistic state estimation. Proven ability to deliver integrated hardware–software solutions for autonomous navigation, teleoperation, and mobile robotics.

EXPERIENCE

Robotics Research Intern — AIST (National Institute of Advanced Industrial Science and Technology)

Tokyo, Japan · Jul 2025 – Jan 2026

- Independently developed a ROS2-based head teleoperation module for a humanoid robot using VR tracking, implementing spatial calibration, Cartesian motion mapping, and axis-dependent scaling.
- Designed trajectory smoothing algorithms and integrated inverse kinematics with joint constraints and acceleration limits to ensure mechanical safety and stable articulation.
- Conducted robustness testing under communication noise and micro-movement disturbances, delivering a reliable head control system for embodied telepresence.

PROJECTS

Autonomous Warehouse Forklift — PuzzleBot (ROS2)

ITESM · Jan 2026 – Jun 2026

- Built a fully autonomous differential-drive forklift with EKF localization fusing wheel odometry, ArUco markers, and QR codes; implemented innovation gating (Mahalanobis distance) and multi-marker median fusion for robust pose estimation.
- Developed a complete perception stack: YOLOv8 logo detection with vote-accumulation for bay-door mapping, WeChat QR decoder with solvePnPGeneric IPPE pose estimation, and ArUco-based relocalization across a 12-marker map.
- Implemented A* global planning with DWA local obstacle avoidance on live LiDAR data; deployed a distributed ROS2 architecture across NVIDIA Jetson (hardware drivers) and laptop (perception + nav), with QoS-tuned topics and UDP-only DDS transport.

Intelligent Autonomous Guidance System — John Deere Challenge

ITESM · Jul 2024 – Nov 2024

- Developed navigation algorithms for a small-scale autonomous tractor using IMU and encoder data, integrating SPI, CAN, UART, and I2C protocols across embedded modules.
- Utilized STM32 dual-core processing and real-time feedback mechanisms to optimize system performance.

Navigation System Simulation for Agricultural Tractors — John Deere Challenge

ITESM · Jan 2024 – Jun 2024

- Implemented FreeRTOS-based control on STM32 for real-time monitoring and developed a Python application on Raspberry Pi for UART data acquisition, CSV logging, and live visualization.

6-DOF Robotic Arm — Inverse Kinematics & Manipulation

ITESM · Dec 2024 – Jun 2026

- Designed and programmed a robotic arm with inverse kinematics and dynamic motion control; integrating object-grasping functionalities for manipulation tasks.

EDUCATION

BEng in Robotics and Digital Systems — ITESM (Tec de Monterrey)

Monterrey, México · Jan 2022 – Jun 2026

International Baccalaureate — Colegio Arji

Villahermosa, México · Jan 2019 – Jan 2021

TECHNICAL SKILLS

Languages: C/C++ · Python · MATLAB · Java

Platforms: STM32 · Raspberry Pi · ESP32 · NVIDIA Jetson · Linux

Frameworks: ROS2 · FreeRTOS · OpenCV · NumPy · YOLOv8 · Simulink · Git

Sensors & Protocols: LiDAR · IMU · Camera (RGB/CSI) · SPI · UART · I2C · CAN

Algorithms: EKF · A* · DWA · PnP Pose Estimation · SLAM · Visual Servoing

LANGUAGES

Spanish (Native) · English (C1)